WEEK 5 PROJECT SUMMARY

1. What Was Planned

The Week 5 plan was to move from the Week 4 feature-analysis stage into practical data preparation, feature engineering and baseline model development.

2. What Was Completed

The following activities were implemented:

- ✓ Confirmed the machine-learning target.
- ✓ Defined the binary no-show modelling problem.
- ✓ Examined cancelled appointments.
- ✓ Created the modelling dataset.
- ✓ Assessed duplicate records.
- ✓ Assessed missing values.
- ✓ Investigated categorical variables.
- ✓ Standardised text values.
- ✓ Investigated rare and high-cardinality categories.
- ✓ Investigated potential identifier variables.
- ✓ Checked numerical ranges.
- ✓ Investigated outliers using IQR analysis.
- ✓ Converted date variables.
- ✓ Engineered waiting-time and appointment-timing features.
- ✓ Assessed target distribution.
- ✓ Performed feature/outcome visual analysis.
- ✓ Implemented leakage controls.
- ✓ Defined a train/test strategy.
- ✓ Implemented numerical preprocessing.
- ✓ Implemented OneHotEncoding for categorical variables.
- ✓ Developed a Dummy Classifier baseline.
- ✓ Developed a Logistic Regression baseline.
- ✓ Evaluated the baseline model using classification metrics.
- ✓ Produced a confusion matrix.
- ✓ Examined Logistic Regression coefficients.

3. Key Outcomes

The Week 5 work produced an initial modelling pipeline that can be used as a baseline for future model development.

4. Major Challenges

The main challenges involved data quality, potential outliers, categorical variables, high-cardinality fields, target leakage and determining which information would realistically be available before an appointment.

5. Important Decisions

The primary modelling problem was defined as binary no-show prediction. Cancelled appointments were excluded from the primary model.
Preprocessing was implemented through a Scikit-learn pipeline to reduce the risk of train/test contamination.
Categorical variables are transformed using OneHotEncoding.

6. Changes From Week 4

The Week 4 feature-analysis recommendations were used as the foundation for the Week 5 modelling dataset.
The Week 5 work introduced practical data cleaning, feature engineering, preprocessing and baseline modelling.

7. Cross-Track Collaboration

The Data Analytics track was identified as an important collaboration point because descriptive attendance patterns and KPIs can inform predictive feature selection and model interpretation.

8. Remaining Work

Further work is required to:

- ✓ Compare additional algorithms.
- ✓ Perform cross-validation.
- ✓ Investigate class imbalance.
- ✓ Conduct further feature selection.
- ✓ Evaluate probability thresholds.
- ✓ Investigate fairness.
- ✓ Tune promising models.
- ✓ Validate the model against additional data-quality scenarios.
- ✓ Incorporate Complex Algorithms

9. Week 6 Focus

Week 6 should focus on improving the baseline through model comparison, cross-validation, feature refinement, threshold analysis and deeper model evaluation.